

Custom K8s Distribution

Upstream source with -ckp suffix, PGP-signed by Coreedge.io

Full CAPI Lifecycle

Kamaji hosted control plane, BYOH provisioning, 14-step flow

Dual Infrastructure

Orbiter Baremetal (BMS) + CCS Virtual Machine (CCP)

PLATFORM OVERVIEW

CKP (Coreedge Kubernetes Platform) is a custom Kubernetes distribution built on upstream Kubernetes source by Coreedge.io. CKP binaries are functionally identical to upstream, but versioned, packaged, and **digitally signed by Coreedge.io** for enterprise traceability and supply chain integrity. All supported versions are **CNCF Certified**.

Platform Type	Custom Kubernetes Distribution + Cluster Lifecycle Management
Vendor	Coreedge.io
Certification	CNCF Certified Kubernetes (all supported versions)
Package Signing	PGP-signed by Coreedge.io with mandatory integrity verification
Architecture	Two-layer: Distribution Layer (distro) + Management Layer (CAPI/Compass)
Supported Versions	v1.29.0, v1.30.6, v1.31.2, v1.32.11, v1.33.7, v1.34.3, v1.35.1
Version Suffix	-ckp (e.g., v1.31.2-ckp)

DISTRIBUTION LAYER

CKP Packages (Debian / AMD64)		BYOH Bundle Distribution	
kubeadm	Cluster bootstrapping & lifecycle	Ubuntu 20.04	All K8s versions (v1.29.0+)
kubelet	Primary node agent (pod execution)	RHEL 9	All K8s versions (v1.29.0+)
kubectrl	CLI for Kubernetes API	Format	OCI-compliant image artifacts
kubernetes-cni	Container Network Interface plugins	Delivery	imgpkg tool (version-locked)
cri-tools	CRI diagnostic tools (crictl)	Registry	Coreedge Docker Hub
containerd	Container runtime with CNI support	Images	kube-apiserver, etcd, CoreDNS, pause...

COMPATIBILITY MATRIX

K8s Version	etcd	CoreDNS	Containerd	CRI	Pause	Calico	CNCF
v1.29.0	3.5.12-0	v1.11.1	1.7.0+	v1	3.9	v3.28.2	Yes
v1.30.6	3.5.15-0	v1.11.3	1.7.0+	v1	3.9	v3.28.2	Yes
v1.31.2	3.5.15-0	v1.11.3	1.7.0+	v1	3.10	v3.30.5	Yes
v1.32.11	3.5.15-0	v1.11.3	1.7.0+	v1	3.10	v3.30.5	Yes
v1.33.7	3.5.15-0	v1.11.3	1.7.0+	v1	3.10	v3.30.5	Yes
v1.34.3	3.5.15-0	v1.11.3	1.7.0+	v1	3.10	v3.30.5	Yes
v1.35.1	3.5.15-0	v1.11.3	1.7.0+	v1	3.10	v3.30.5	Yes

MANAGEMENT LAYER (CAPI + COMPASS)			
Cluster API (Core)		v1.7.7	
Kubeadm (Bootstrap + CP)		v1.7.7	
BYOH (Infrastructure)		v0.6.1	
Kamaji (Control Plane)		v0.16.0 — Konnectivity, CoreDNS, KubeProxy, LoadBalancer	
Cert-Manager		v1.15.3 (10-year TLS certificates)	
Compass Platform		Unified UI + REST/gRPC APIs; multi-tenant (domain + project scoped)	
INFRASTRUCTURE PROVIDERS			
Capability	Orbiter Baremetal (BMS)		CCS Virtual Machine (CCP)
Provisioning	Cloud-init templates		Automated VM lifecycle
Networking	Physical server allocation		Neutron integration, security groups
Autoscaling	Not supported		Karpenter (CPU-based)
Architecture	AMD64		AMD64 + ARM64
OS Support	Ubuntu 20.04, RHEL 9		Ubuntu 20.04, RHEL 9
Best For	Dedicated, high-performance		Flexible, multi-arch, elastic scaling
NETWORKING, STORAGE & BACKUP			
Networking & Storage		Backup & Autoscaling	
Pod Network	Configurable CIDR	Backup	Velero (S3), project-level isolation
Service Network	Configurable CIDR	Autoscaler	Karpenter (CCS VM only)
CNI (CAPI)	Cilium (eBPF-based)	Scaling Limits	CPU-based
CNI (Compass)	Calico v3.28.2 / v3.30.5	Storage Class	ckp-block (Delete reclaim, expandable)
Storage	Ceph (ckp-block), OpenEBS (standalone)	Deployment	Helm chart
SECURITY, COMPLIANCE & PREREQUISITES			
Security & Compliance		System Prerequisites	
Package Signing	PGP by Coredge.io; mandatory	Supported OS	Ubuntu 20.04, 22.04, RHEL 9
Image Registry	Coredge Docker Hub	Container Runtime	Containerd (v1.6.14+)
Host Auth	mTLS + Pre-Shared Key (PSK)	OCI Runtime	runc (v1.1.3 – v1.1.10)
TLS Certificates	10-year validity; Cert-Manager	CRI / CNI	crictl v1.27.0 / CNI v1.1.2–v1.3.0
Build Security	Hardened base; non-root; CNCF Certified	Kernel	overlay, br_netfilter, swap off
CLUSTER LIFECYCLE (14-STEP END-TO-END)			
Phase	Steps	Key Components	
Provisioning	1–4	Request cluster → Resolve provider → Provision hosts → Register (mTLS)	
Bootstrap	5–8	Approve hosts → Assign group → CAPI resources → Reconcilers	
Cluster Ready	9–11	Kamaji CP up → Workers join → Ready state	
Addons & Scaling	12–14	CKP Storage + Cilium + Velero + Karpenter + 10-year TLS certs	

Coredge.io — Coredge Kubernetes Platform (CKP) — All content sourced from CKP Master Document